

ADAPTIVE LEARNING SYSTEM FOR THAI MUSIC EDUCATION USING AI AND AUGMENTED REALITY

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ABSTRACT. *This paper proposes THAI-MUSE (Thai Music Understanding and Skill Enhancement), an innovative adaptive learning system for Thai traditional music education that integrates artificial intelligence and augmented reality technologies. The system addresses critical challenges in traditional music instruction, including limited access to expert teachers and declining student engagement. THAI-MUSE employs a novel approach combining deep learning for performance analysis with reinforcement learning for personalized content delivery, all surfaced through an augmented reality interface that provides real-time feedback. The proposed architecture features specialized neural network models for Thai musical elements and a culturally sensitive content adaptation algorithm. This paper presents the theoretical framework, system architecture, and technical specifications without implementation, focusing on how AI-driven adaptive learning can support the preservation and transmission of traditional musical knowledge while respecting cultural authenticity.*

Keywords: Adaptive learning, Thai music, Music education, Artificial intelligence, Augmented reality, Cultural preservation

1. Introduction. Thai traditional music embodies a rich cultural heritage characterized by unique tonal systems, distinctive instruments, and complex performance practices [1]. Despite its cultural significance, traditional music education in Thailand faces several critical challenges:

- Declining availability of master musicians qualified to teach
- Limited access to quality instruction outside major urban centers
- Lack of explicit standardized teaching methodologies
- Decreasing interest among younger generations [2]

Digital technologies have demonstrated potential for addressing similar challenges in music education globally [3]. However, the unique characteristics of Thai music – including its modal structures, improvisation practices, and specialized instrumental techniques – require tailored technological approaches beyond conventional digital music education systems [4].

This paper introduces THAI-MUSE, an artificial intelligence-driven adaptive learning system enhanced with augmented reality capabilities designed specifically for Thai traditional music education. The system aims to

- 1) Analyze student performances using deep learning models specialized for Thai musical elements

- 2) Adapt instructional contents based on individual learning patterns
- 3) Provide real-time feedback through augmented reality visualization
- 4) Preserve authentic teaching methodologies while improving accessibility

Our approach integrates cultural preservation with technological innovation, creating a framework that could potentially be adapted for other traditional music systems facing similar preservation challenges.

This research makes several novel contributions. 1) We propose the first comprehensive adaptive learning architecture for traditional non-Western music education, integrating performance analysis, knowledge modeling, and immersive feedback. 2) We develop custom neural network architectures for Thai musical elements including micro-tonal inflections, modal structures (*khroongsang phleng*), and traditional ornamentations (*uean*, *sabat*, *prakhop siang*). 3) We introduce a hybrid content adaptation approach combining Deep Q-Network reinforcement learning with traditional Thai pedagogical principles. 4) We propose an AR interface design tailored to Thai instrumental techniques. 5) We establish a methodological framework for developing AI-driven educational systems for endangered cultural practices – applicable to other traditional music systems worldwide. These contributions address a significant research gap at the intersection of adaptive learning, cultural preservation, and AI-enhanced music education.

The remainder of this paper is organized as follows. Section 2 reviews related work in AI-based music education, augmented reality applications in music instruction, and existing technologies for traditional music preservation, identifying critical gaps that motivate our research. Section 3 presents the THAI-MUSE system architecture in detail, describing the Performance Analysis Engine (Section 3.2), Knowledge Modeling System (Section 3.3), Content Adaptation Engine (Section 3.4), and Augmented Reality Interface (Section 3.5). Section 4 provides technical implementation specifications including audio processing frameworks, AI model specifications, AR requirements, and data requirements. Section 5 discusses potential educational impact, cultural preservation considerations, technical challenges, and future research directions. Finally, Section 6 concludes with a summary of contributions and implications for music education technology and cultural heritage preservation.

2. Related Work.

2.1. AI in music education. Applications of artificial intelligence in music education have proliferated in recent years, with various systems demonstrating the efficacy of machine learning for performance analysis and instructional adaptation. Notable examples include piano performance assessment systems using convolutional neural networks [5] and violin tutoring systems employing hidden Markov models for bowing technique evaluation [6].

For non-Western musical traditions, AI applications remain more limited. Systems for Indian classical music have focused primarily on raga classification rather than performance instruction [7], while approaches to Chinese traditional music have emphasized notation and theoretical aspects over performance [8].

2.2. Augmented reality in music instruction. Augmented reality has shown promise for visualizing abstract musical concepts and providing real-time guidance. Several systems have demonstrated AR applications for Western instruments, including piano keyboard visualization [9] and guitar fingering guidance [10].

For traditional music specifically, [11] developed an AR application for Indonesian traditional string instruments that visualized finger positions, showing improved learning outcomes in laboratory settings. However, existing AR applications for traditional music education typically lack adaptive learning capabilities and sophisticated performance analysis.

2.3. Traditional music preservation technologies. Digital approaches specifically targeting traditional Thai music education relatively remain underdeveloped. Existing technologies have focused primarily on static content delivery, including notation software [12] and basic instructional applications [13]. None have effectively integrated performance analysis, adaptive learning, and immersive feedback – elements critical for comprehensive music instruction.

2.4. Research gaps and motivation. While the reviewed literature demonstrates progress in AI-based music education, AR applications, and digital cultural preservation, several critical gaps motivate this research.

Gap 1: Existing AI music education systems [5-8] focus predominantly on Western traditions. Thai music has unique modal structures, micro-tonal inflections, and improvisation practices requiring specialized approaches. No system integrates performance analysis, personalized adaptation, and immersive feedback for traditional Thai music.

Gap 2: AR applications for music [9-11] typically lack sophisticated learner modeling, while AI-based systems lack immersive interfaces supporting embodied learning. Integration of deep learning analysis with reinforcement learning adaptation, delivered through culturally tailored AR, remains unexplored.

Gap 3: Standard audio analysis models trained on Western music perform poorly on Thai music characteristics [4]. Existing models do not recognize Thai-specific elements (*uean*, *sabat*, *khongsang phleng*), and knowledge modeling approaches [5,6] have not been adapted to traditional Thai pedagogical sequences.

Gap 4: Research lacks frameworks for AI-enhanced educational systems balancing innovation with cultural preservation. Technologies for traditional music [12,13] focus on static delivery without addressing how adaptive systems can support traditional knowledge transmission.

THAI-MUSE addresses these gaps by developing specialized AI models, integrating adaptive learning with AR delivery, and establishing a culturally grounded framework applicable beyond Thai music.

3. System Architecture.

3.1. System overview. THAI-MUSE integrates four primary components to create an adaptive learning environment for traditional Thai music education:

- 1) **Performance Analysis Engine:** Processes audio and visual input to evaluate student performances
- 2) **Knowledge Modeling System:** Maintains representations of student knowledge and skill proficiency
- 3) **Content Adaptation Engine:** Selects and modifies instructional content based on student needs
- 4) **Augmented Reality Interface:** Delivers instruction and feedback through immersive visualization

Figure 1 illustrates the system architecture and information flow between components.

The system operates through a continuous feedback loop: student performances are captured and analyzed, the results update the student's knowledge model, the adaptation engine selects appropriate content, and the AR interface delivers instruction and feedback. Each component is described in detail below.

3.2. Performance Analysis Engine. The Performance Analysis Engine employs multiple AI approaches to evaluate student performances comprehensively. The engine processes both audio input (for musical analysis) and visual input (for technique assessment).

Audio Analysis Component. The audio analysis pipeline consists of three stages.

- 1) **Preprocessing:** Audio signals undergo normalization, noise reduction, and segmentation into analysis frames. Thai music's unique timbral characteristics require specialized preprocessing parameters.
- 2) **Feature Extraction:** The system extracts multiple feature sets:
 - Mel-Frequency Cepstral Coefficients (MFCCs) for timbral analysis
 - Pitch class profiles for tonal analysis
 - Onset detection functions for rhythmic analysis
 - Custom features for Thai-specific ornamentations, e.g., *uean* (melisma), *sabat* (three-syllable flip), and *prakhop siang* (tonal articulation)
- 3) **Deep Learning Classification:** A specialized neural network architecture processes these features to identify performance elements and detect errors. Table 1 summarizes the model design.

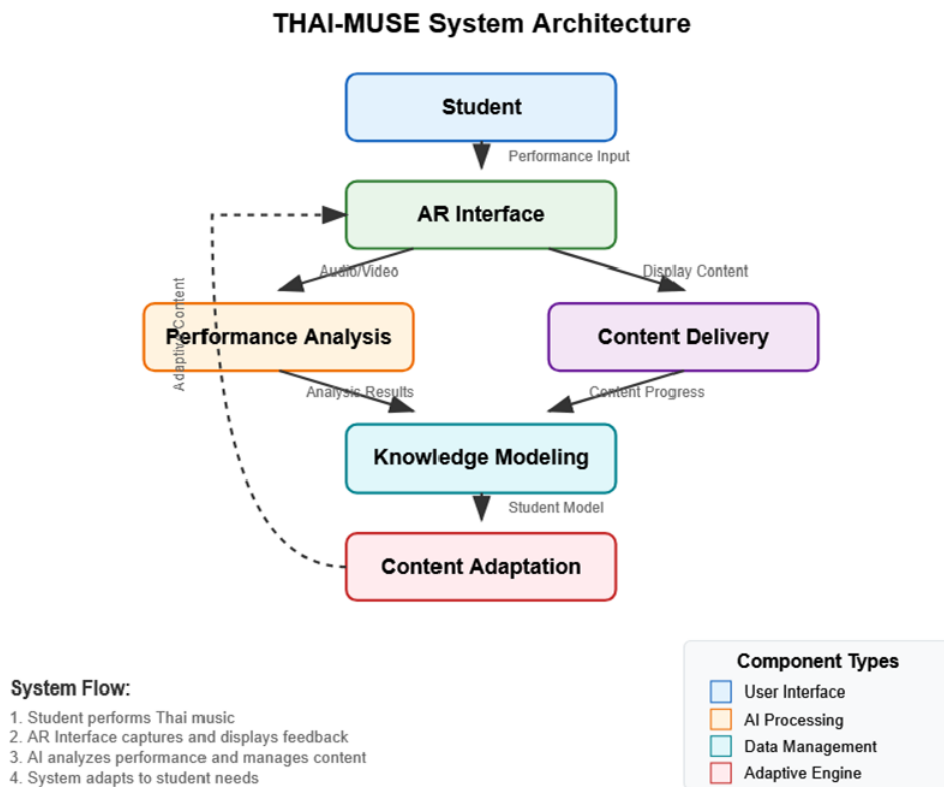


FIGURE 1. (color online) THAI-MUSE system architecture

TABLE 1. Audio analysis model design

Layer type	Configuration	Purpose
Input	Multi-channel feature maps	Process multiple acoustic feature types
Convolutional blocks	3 blocks with (64, 128, 256) filters	Extract hierarchical patterns
Bi-directional LSTM	2 layers, 128 units each	Capture temporal dependencies
Attention mechanism	Self-attention over time	Focus on salient musical events
Classification head	Multi-task outputs	Separate evaluations for pitch, rhythm, and expression

The model is designed to recognize Thai musical characteristics including

- Modal structures (*khrongsang phleng*) specific to Thai music
- Micro-tonal inflections essential to traditional performance practice
- Complex rhythmic patterns (*krasuan changwa*) characteristic of different musical forms

The visual analysis component tracks hand and body positions essential for proper instrumental technique. For instruments like the *ranat ek* (treble wooden xylophone) or the *khong wong yai* (large circle gong chime), correct mallet handling significantly impacts sound quality.

The visual system employs a pose estimation model based on the architecture proposed by [14], with specialized hand keypoint detection crucial for Thai instrumental technique. The model is designed to identify

- Correct mallet grip and wrist position for melodic percussion instruments
- Proper finger placement for stringed instruments (e.g., *so duang*, and *chakhe*)
- Body posture appropriate for each instrument

3.3. Knowledge Modeling System. The Knowledge Modeling System maintains a comprehensive representation of each student's proficiency across multiple dimensions of musical skill. Rather than a simplistic correct/incorrect binary, the system models knowledge as probabilistic mastery of hierarchical skill components.

The knowledge model employs a Bayesian network structure [15,16] where

- Nodes represent specific skills (e.g., basic striking technique, and rhythmic pattern execution)
- Edges represent prerequisite relationships between skills
- Node probabilities represent estimated skill mastery (0-1)
- Evidence from performance updates these probabilities

Figure 2 illustrates a simplified segment of the skill network for *ranat ek* performance.

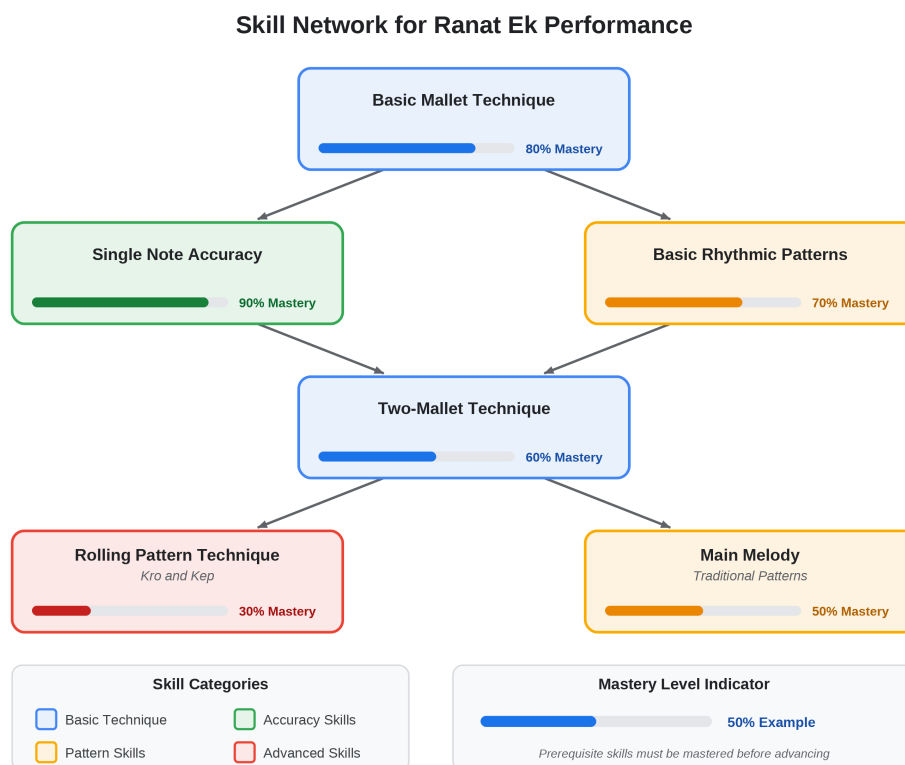


FIGURE 2. (color online) Partial skill network for *ranat ek* performance

The knowledge model incorporates

- Separate tracking for different instruments and musical forms
- Confidence estimates for each skill probability
- Temporal models of skill acquisition and decay
- Relationships between theoretical knowledge and performance skills

This multidimensional modeling allows the system to target instruction precisely to student needs, addressing specific weaknesses while recognizing strengths.

3.4. Content Adaptation Engine. The Content Adaptation Engine determines what content to present to students based on their current knowledge state, learning history, and educational objectives. The system conceptualizes learning as a Markov Decision Process (MDP) and employs reinforcement learning to optimize instructional sequencing.

In this formulation:

- States (S) represent student knowledge profiles
- Actions (A) represent available learning activities
- Transitions (T) model how knowledge changes after completing activities
- Rewards (R) reflect immediate learning gains
- Policy (π) determines the optimal activity selection strategy

The adaptation engine implements a Deep Q-Network (DQN) approach that approximates the Q-function using a neural network with the architecture specified in Table 2.

TABLE 2. DQN architecture for content adaptation

Layer	Type	Configuration	Activation
Input	Dense	Student knowledge vector (157 dimensions)	–
Hidden 1	Dense	128 units	ReLU
Hidden 2	Dense	64 units	ReLU
Output	Dense	Action values for learning activities	Linear

Q-Learning Implementation Details. The system employs Q-learning [17], a model-free reinforcement learning algorithm, to learn optimal content selection policies. The Q-function $Q(s, a)$ represents expected cumulative reward of taking action a in state s . The update rule is

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where $\alpha = 0.001$ (learning rate), $\gamma = 0.95$ (discount factor), r is immediate reward, s' is next state, and $\max_{a'} Q(s', a')$ is the maximum Q-value across actions in the next state.

State and Action Representation: The state space S is a 157-dimensional vector corresponding to skill mastery probabilities in the Bayesian knowledge network (Section 3.3), augmented with temporal decay, engagement metrics, and performance trends. The action space A consists of discrete learning activities characterized by target skills, difficulty level (1-5), activity type, and cultural context integration.

Reward Function: The reward $r(s, a, s')$ incorporates skill mastery gains (ΔP), engagement (E), zone of proximal development alignment (Z), and frustration penalty (F): $r(s, a, s') = 0.5 \cdot \Delta P + 0.2 \cdot E + 0.2 \cdot Z - 0.1 \cdot F$.

Training Mechanisms: The DQN employs experience replay (100,000 capacity buffer, batch size 32) and dual networks (target network updated every 1,000 steps) for stable learning [17]. Exploration uses ε -greedy strategy with ε decaying from 0.3 to 0.05 (decay rate 0.995).

Cultural Pedagogy Integration: The system balances algorithmic optimization with traditional teaching practices through 1) hard constraints preventing access to advanced content without prerequisite mastery, 2) action probability weighting favoring traditional

pedagogical sequences, and 3) incorporation of teacher overrides into training data. This hybrid approach ensures the RL policy respects authentic cultural knowledge transmission while optimizing learning efficiency.

The content library is organized hierarchically, covering fundamental skills through advanced repertoire. Each learning activity is tagged with

- Prerequisite skills
- Target learning outcomes
- Difficulty level (1-5)
- Engagement mechanisms
- Cultural context elements

The adaptation process incorporates several key principles:

- 1) **Zone of Proximal Development:** Selecting content challenging enough to promote learning but not so difficult as to cause frustration
- 2) **Spacing Effect:** Scheduling reviews of previously learned material at optimal intervals
- 3) **Interleaving:** Mixing different skill types and content areas to improve retention
- 4) **Cultural Progression:** Following traditional Thai music pedagogical sequences

The adaptation algorithm itself employs a hybrid approach combining reinforcement learning with domain-specific heuristics drawn from conventional Thai music pedagogy.

3.5. Augmented Reality Interface. The Augmented Reality Interface serves as the primary means of interaction between the student and the system. It overlays instructional content and performance feedback directly onto physical instruments through a tablet or smartphone camera.

The AR interface includes several key features.

- 1) **Visual Guidance:** Animated overlays demonstrate correct hand movements and playing positions. For the *ranat ek*, this includes mallet trajectory visualization showing optimal striking positions and angles.
- 2) **Real-time Feedback:** Color-coded highlights indicate performance accuracy, with green showing correct notes/timing and red indicating errors.
- 3) **Traditional Notation Integration:** The system displays Thai alphabetical notation (*not thai*) alongside the instrument, with synchronized highlighting as the piece progresses.
- 4) **Performance Visualization:** Waveforms and spectrograms provide visual representations of acoustic output, helping students understand the relationship between technique and sound.

Figure 3 illustrates the conceptual design of the AR interface for *ranat ek* instruction.

The interface adapts based on student proficiency, gradually reducing scaffolding as skills develop. For beginners, it provides explicit visual guidance before performance; for intermediate students, it shifts toward post-performance feedback; for advanced students, it focuses on subtle aspects of expression and stylistic authenticity.

4. Technical Implementation Specifications.

4.1. Audio processing framework. The audio processing pipeline requires specific technical implementations to effectively analyze Thai musical characteristics:

- **Sampling Rate:** 44.1 kHz to capture the high-frequency components of Thai percussion instruments
- **Feature Extraction:** 23-dimensional MFCCs with Thai-specific filterbank configurations
- **Windowing:** 25 ms frames with 10 ms overlap for fine-grained analysis
- **Model Training Approach:** Transfer learning from Western music models with fine-tuning on Thai music corpus

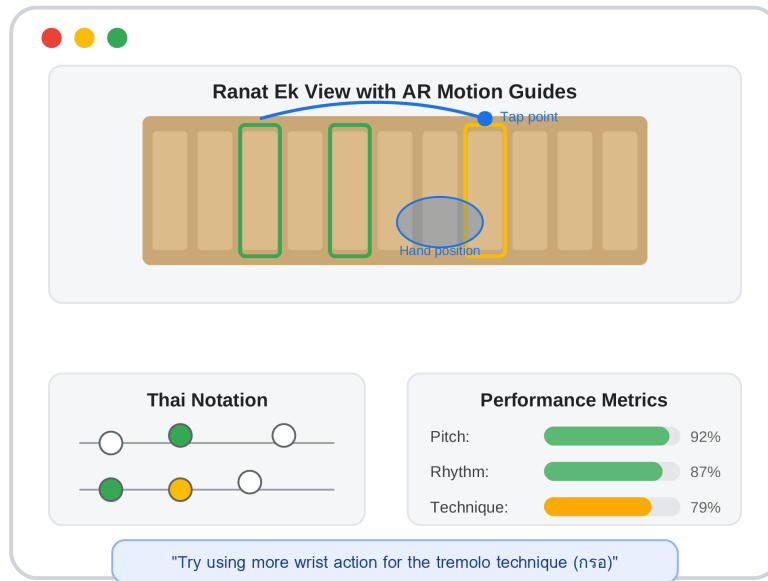


FIGURE 3. AR interface concept for *ranat ek* instruction

4.2. **AI model specifications.** The core AI models require substantial computational resources during training but are optimized for efficient inference on consumer hardware:

- **Performance Analysis CNN:** 12.7 M parameters, optimized for mobile inference (8-bit quantization)
- **Knowledge Tracing Model:** Bayesian network with 157 nodes for comprehensive skill modeling
- **Content Adaptation DQN:** Updated asynchronously to optimize content selection

4.3. **AR implementation requirements.** The AR system has specific technical requirements for effective operation:

- **Tracking Method:** Marker-based with marker-less fallback using ARCore/ARKit
- **Hardware Requirements:** Tablet/smartphone with minimum 4GB RAM, integrated camera
- **Rendering Pipeline:** OpenGL ES 3.0 for cross-platform compatibility
- **Latency Requirements:** < 50 ms processing latency for real-time feedback

4.4. **Data requirements.** The system requires substantial data resources for effective operation:

- **Audio Corpus:** Professionally recorded examples of correct performance (minimum 500 recordings)
- **Expert Annotations:** Performance evaluations from master musicians (minimum 3 per recording)
- **Learning Content:** Structured lessons covering fundamental through advanced techniques
- **Cultural Context:** Historical and cultural materials to provide context beyond technique

5. Discussion.

5.1. **Potential educational impact.** THAI-MUSE has the potential to address several critical challenges in Thai music education.

Accessibility: By providing quality instruction without requiring constant access to master teachers, the system could significantly extend educational reach to underserved regions.

Standardization: The system offers consistent instruction while still accommodating regional stylistic variations, potentially helping to standardize fundamental aspects of technique.

Engagement: The integration of familiar technology may increase appeal among younger generations, potentially revitalizing interest in traditional forms.

Efficiency: Adaptive learning potentially accelerates skill acquisition for foundational techniques, allowing human teachers to focus on advanced stylistic elements.

5.2. Cultural preservation considerations. A critical concern in developing technology for cultural heritage education is maintaining authenticity while leveraging modern approaches. THAI-MUSE addresses this through

- 1) **Expert Knowledge Integration:** The system's models are designed to incorporate expert knowledge from master musicians rather than imposing computational simplifications.
- 2) **Stylistic Flexibility:** The performance analysis models accommodate legitimate stylistic variations rather than enforcing rigid interpretations.
- 3) **Cultural Context:** Instructional content integrates historical and cultural information alongside technical instruction.
- 4) **Supplementary Role:** The system is positioned as a supplement to, rather than replacement for, traditional human instruction.

The potential exists for technology-mediated instruction to alter traditional knowledge transmission patterns. While this raises legitimate concerns, it also presents opportunities for preservation in contexts where traditional methods are already declining.

5.3. Technical challenges. Several significant technical challenges must be addressed for successful implementation.

Acoustic Complexity: Thai musical instruments produce complex harmonic structures and micro-tonal variations that challenge conventional audio analysis approaches.

Cultural Modeling: Formalizing tacit cultural knowledge into computational models requires extensive collaboration with master musicians and ethnomusicologists.

Evaluation Metrics: Defining appropriate evaluation metrics that respect cultural values rather than imposing Western music assessment frameworks.

AR Stability: Maintaining stable AR overlays under various lighting conditions and during rapid movement requires robust tracking solutions.

5.4. Future directions. While this paper presents theoretical framework and system design, several promising directions for future research emerge:

- 1) **Multimodal Learning:** Integrating haptic feedback alongside visual and auditory instruction could further enhance embodied learning;
- 2) **Collaborative Features:** Extending the system to support ensemble practice through networked synchronization;
- 3) **Longitudinal Studies:** Designing studies to assess long-term impacts on both skill acquisition and cultural understanding.

6. Conclusion. This paper has presented THAI-MUSE, an artificial intelligence-driven adaptive learning system enhanced with augmented reality for Thai traditional music education. The proposed system integrates specialized performance analysis, knowledge modeling, content adaptation, and immersive visualization to create personalized learning experiences that respect cultural authenticity.

By addressing the specific challenges of Thai music education through targeted technological innovation, THAI-MUSE offers a promising approach to preserving and transmitting cultural heritage in increasingly digital educational contexts. While not intended

to replace traditional instruction, such systems can potentially extend educational reach and revitalize interest in traditional forms.

The framework presented here contributes to both music education technology and cultural heritage preservation by demonstrating how AI and AR can be applied in culturally sensitive ways that support rather than supplant traditional knowledge systems. This approach may serve as a model for similar efforts in other cultural traditions facing preservation challenges in the digital age.

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